Date: _____ Day: _____

# Section no 2
# Probability

## Uncertainity Due to Randomness:

Uncertainity: outcome of an experiment is not known in advance.

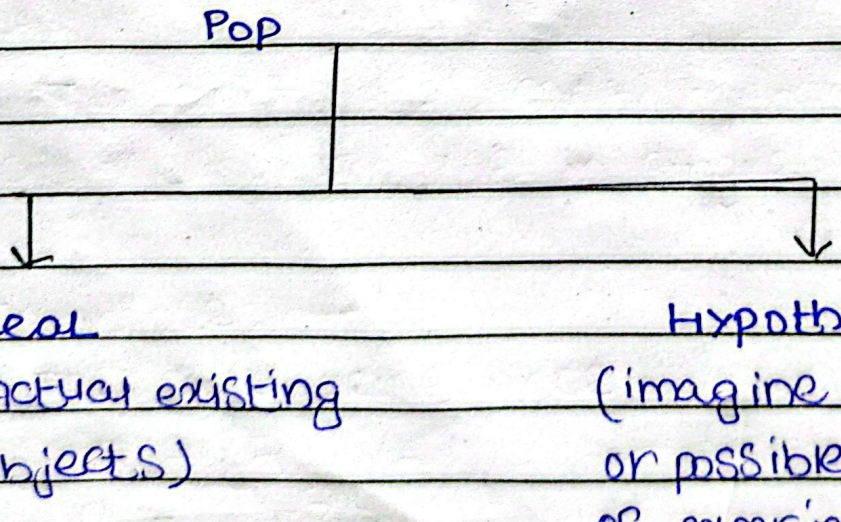
marks = f(Std Hrs, Travel Hrs)

we can control — we can manage

But:

accident: these are random events. they may effect results

Probability: study that studies the chance of an event happening.



Date: Day:

Result

Predictable

If predicable then Deterministic.

Unpredictable

If unpredictable event when repeated Large no of time under same condition then Random.

Event: any set of outcome from sample space.

Sample space: set of all possible outcomes of experiment.

outcome: single, unique possible result of experiment eg H or t

Trial: Performing an experiment once e.g Rolling die one time.

NULL Event: impossible event e.g Getting 5 & 6 on single Die.

Sure Event: event that must happen e.g two coins are tossed

$S = \{HH, HT, TH, TT\}$ → sure event

Date: _____ Day: _____

**Simple Event:** that contain one outcome.

**Compound Event:** contain two or more outcome.

$S = \{HH, HT, TH, TT\}$

$\emptyset$ → Null event

$\{HH\}, \{HT\}, \{TH\}, \{TT\}$ → Simple event

$\{HH, HT\}, \{HH, TH\}, \{HH, TT\}$
$\{HT, TH\}, \{HT, TT\}$
$\{TH, TT\}$
$\{HH, HT, TH\}, \{HH, TH, TT\}$
$\{HT, TH, TT\}$
→ Compound Event

$\{HH, HT, TH, TT\}$ → Sure event and Compound.

contain all possible outcome & must happen.

**Mutually Exclusive:**

Two events that can't happen at the same time in single trial.

**Example:**

Tossing a coine once.

$A = \{H\}$

$B = \{T\}$

Date: _____ Day: _____

We can't get both Head and Tail in one toss.

$$A \cap B = \emptyset$$

**Exhaustive Event:**

Two or more events are exhaustive if together they cover ALL possible outcomes of an experiment.

Example: Rolling Die

$A = \{2, 4, 6\}$

$B = \{1, 3, 5\}$

Together

$$A \cup B = \{1, 2, 3, 4, 5, 6\}.$$

Now A and B are exhaustive coz they cover every possible outcome.

**Equally Likely Event:**

Events that have same/Fair chance of occuring.

Example: Tossing a fair coin both H and T have equally chance.

Date: Day:

# Probability

## Objective

- Measure by some well define methods
- Prior knowledge matters
- Based on Logic & Reasoning

## Subjective:

Based on someone personal Judgement or belief

### Classical Definition

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(\text{event}) = \frac{\text{No of Favorable Outcomes}}{\text{No of Possible Outcomes}}$$

### Relative Freq.

### Axiomatic Definition.

**Permutation:-**

is an arrangements of objects in which order matters

**Example:**

Arrange A & B

Date: _____ Day: _____

- Possible Arrangements: AB, BA
- Here AB and BA are different.

Formula:

$$^nP_r = \frac{n!}{(n-r)!}$$

**Combination:**

Selection of object in which order does't matter.

Example:

- Select 2 students A and B
- Possible selection: AB
- AB and BA represent same selection.

Formula:

$$^nC_r = \frac{n!}{r!\,(n-r)!}$$

## Addition Law For Probability:

Find Probab. that atleast one of two events (A or B) will occur.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- $P(A \cup B)$: Prob that either event A or Event B occurs
- $P(A)$: Prob of A
- $P(B)$: Prob of B
- $P(A \cap B)$: Prob of A and B occuring at same time.

If two event can't happen at same time.

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$ becomes 0

Date: $P(A \cup B) = P(A) + P(B)$ Day:

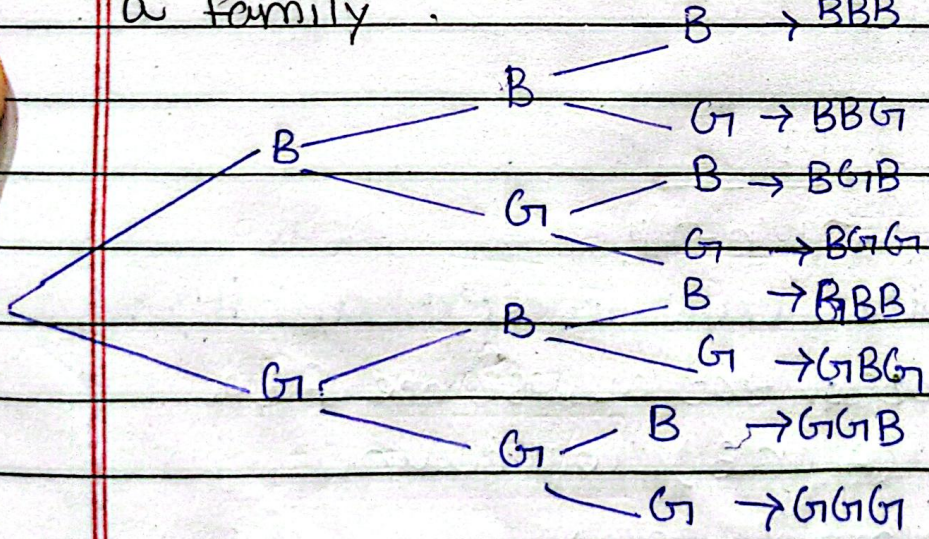
## Example: 4 - 1 (Bluman Pg 183)

Find Sample space For rolling two Dice.

| | Die 2 | | | | | |
|---|---|---|---|---|---|---|
| Die 1 | 1 | 2 | 3 | 4 | 5 | 6 |
| 1 | (1,1) | (1,2) | (1,3) | (1,4) | (1,5) | (1,6) |
| 2 | (2,1) | (2,2) | (2,3) | (2,4) | (2,5) | (2,6) |
| 3 | (3,1) | (3,2) | (3,3) | (3,4) | (3,5) | (3,6) |
| 4 | (4,1) | (4,2) | (4,3) | (4,4) | (4,5) | (4,6) |
| 5 | (5,1) | (5,2) | (5,3) | (5,4) | (5,5) | (5,6) |
| 6 | (6,1) | (6,2) | (6,3) | (6,4) | (6,5) | (6,6) |

## Example: Bluman Pg # (185) - 4 - 4

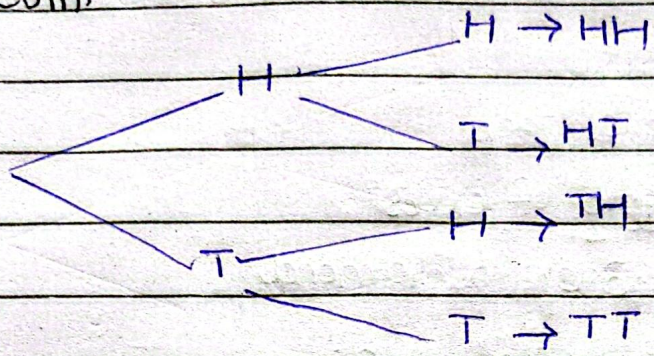
Use a tree diagram to find sample space for gender of three children in a family.



Date: _____ Day: _____

## Example:

Find Sample space For Tossing two coin.




## Example: 4-5 Bluman Pg # 187

Find a Probability of getting black 10 when drawing from a deck.

one black 10 is in club = 1

one black 10 is in Spade = 2

Let P(black 10)

$$P(\text{black 10}) = \frac{2}{52} = 0.038$$

## Example: 4-6 Bluman Pg # 187.

If family has three children find two of three children are girls.

Let event E: (two of childre three are girl)

$$P(E) = \frac{3}{8} = 0.375$$

Date: _____ Day: _____

## Example: 4-7 (Bluman Pg # 187)

Card is drawn from an ordinary deck; Find these probabilities.

a) of getting a Jack
b) of getting 6 of clubs
c) of getting 3 or a diamond
d) of getting a 3 or a 6.

### a

Given Data :-

Total card = 52 = n(S)

Select = 1

Total Jack = 4

Let event E: (of getting a Jack)

$$P(E) = \frac{n(E)}{n(S)} = \frac{\binom{4}{1}\binom{48}{0}}{\binom{52}{1}} = \frac{(4)(1)}{52}$$

$$P(E) = 0.077$$

$$\boxed{P(E) = 7.7\%}$$

Date: Day:

## b-

Data:

Total card = 52

Select = 1

6 of Club = 1

Let event : G1 (of getting 6 of clubs)

$$P(G_1) = \frac{n(G_1)}{n(S)} = \frac{\binom{1}{1}\binom{51}{0}}{\binom{52}{1}} = \frac{(1)(1)}{52}$$

$P(G_1) = 0.019$

$\boxed{P(G_1) = 1.9\%}$

## c-

of Getting 3 or a Diamond:

Data

total card = 52

Select = 1

Total 3 card = 4

Total Diamond = 13

getting 3 of Diamond = 1

$\therefore P(A \cup B) = P(A) + P(B) - P(A \cap B)$

P(A) : Let event A: (getting 3)

$$P(A) = \frac{n(A)}{n(S)} = \frac{\binom{4}{1}\binom{48}{0}}{\binom{52}{1}} = \frac{(4)(1)}{52}$$

Date: Day:

$P(A) = \frac{4}{52} = 0.077$

Let event B: (getting a Diamond)

$P(B) = \frac{n(B)}{n(S)} = \frac{\binom{13}{1}\binom{39}{0}}{\binom{52}{1}} = \frac{(13)(1)}{52}$

$P(B) = \frac{13}{52} = 0.25$

Let event $(A \cap B)$: (getting 3 of Diamond)

$P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{\binom{1}{1}\binom{51}{0}}{\binom{52}{1}} = \frac{(1)(1)}{52}$

$P(A \cap B) = \frac{1}{52} = 0.019$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(A \cup B) = 0.077 + 0.25 - 0.019$

$P(A \cup B) = 0.308$

$\boxed{P(A \cup B) = 30.8\ \%}$

d-

of getting 3 or a 6.

Total card = 52

Select = 1

3 card = 4

6 card = 4

3 and 6 at same time = 0

Date: Day:

$\therefore P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(A \cup B) = P(A) + P(B) - 0$

$P(A \cup B) = P(A) + P(B)$

Let event A: (of getting 3)

calculated is C part.

$P(A) = 0.0777$ or $\frac{4}{52}$

Let event B: (of getting 6)

$P(B) = \frac{n(B)}{n(S)} = \frac{\binom{4}{1}\binom{48}{0}}{52} = \frac{(4)(1)}{52}$

$P(B) = \frac{4}{52} = 0.0777$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(A \cup B) = \frac{4}{52} + \frac{4}{52} - 0$

$P(A \cup B) = \frac{8}{52} = \frac{2}{13} = 0.154$

$\boxed{P(A \cup B) = 15.4\ \%}$

Date: _____ Day: _____

## Example:

A committee of 3 persons is to be selected randomly from a group of 5 men & 3 women. Find

a) Probability that committee consist of 1 men & women.

b) Prob that committee consist ~~of~~ 1 of all women.

Given Data:

Total members = 8 = n(S)

Select = 3 Men = 5

Women = 3

a:

Let event A: (1 men & Women)

$$P(A) = \frac{n(A)}{n(S)} = \frac{\binom{5}{1}\binom{3}{2}}{\binom{8}{3}} = \frac{(5)(3)}{56}$$

$$P(A) = \frac{15}{56} = 0.268$$

$$P(A) = 26.8\ \%$$

Date: _____ Day: _____

b-

Let event B: (ALL women)

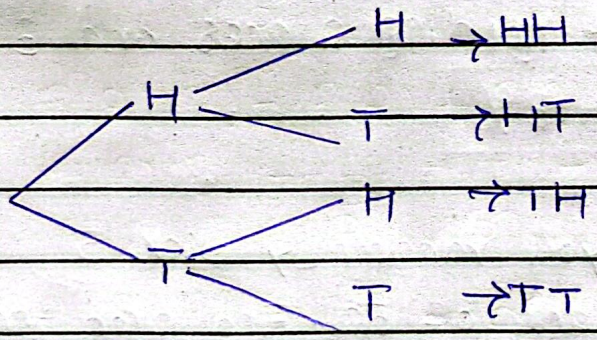
$$P(B) = \frac{n(B)}{n(S)} = \frac{\binom{3}{3}\binom{5}{0}}{\binom{8}{3}} = \frac{(1)(1)}{56}$$

$$P(A) = \frac{1}{56} = 0.018$$

$$\boxed{P(A) = 1.8\ \%}$$

**Example:**

Two Fair coins are tossed what is prob of getting 1 Head & 1 Tail:




Let A: (Prob of 1 Head & 1 Tail)

$$P(A) = \frac{n(A)}{n(S)} = \frac{2}{4} = 0.5$$

$$\boxed{P(A) = 50\ \%}$$

Date: _____ Day: _____

## Example:

A coin is toss twice what is prob that atLeast one Head occur.

$S = \{HH, HT, TH, TT\}$.

Let event A: (atLeast one head)

$$P(A) = \frac{n(A)}{n(S)} = \frac{3}{4} = 0.75$$

$$\boxed{P(A) = 75\%}$$

## Example:

In a Poker Hand consisting of 5 cards find Prob of holding 2 aces and 3 jacks.

or:

eg ask if you randomly pick 5 cards from deck of 52. Find prob that exactly 2 cards are aces and remaining 3 cards are jacks.

Given Data:

Total card = 52 Total Aces = 4

select = 5 Total jacks = 4

Date: _____ Day: _____

Let event A: (3 card are Jack & 3 are Ace)

$$P(A) = \frac{n(A)}{n(S)} = \frac{\binom{4}{2}\binom{4}{3}}{\binom{52}{5}} = \frac{(6)(4)}{2598960}$$

$$P(A) = \frac{24}{2598960} = 0.00000923$$

$$\boxed{P(A) = 0.000923\ \%}$$

**Example:**

A bag contains 6 red balls, 5 yellows balls and 3 green balls. A ball is drawn at random. Find prob.

a) Gree b) Not yellow
c) red or yellow.

Given Data:

Toty balls = 14
Red BALLS = 6 Select = 1
Yellow BALLS = 5
Green BALLS = 3

a-

Let even G: P(Green)

Date: _____ Day: _____

$$P(G) = \frac{n(G)}{n(S)} = \frac{\binom{3}{1}\binom{6}{0}\binom{5}{0}}{\binom{14}{1}} = \frac{(3)(1)(1)}{14}$$

$$P(G) = 0.214$$

$$\boxed{P(G) = 12.4\%}$$

b-

Let event F: P(not yellow)

$$P(F) = \frac{n(F)}{n(S)} = \frac{\binom{3}{1}\binom{6}{0}\binom{5}{0} + \binom{3}{0}\binom{6}{1}\binom{5}{0}}{\binom{14}{1}}$$

$$P(A) = \frac{(3)(1)(1) + (1)(6)(1)}{14} = \frac{9}{14}$$

$$P(A) = 0.643$$

$$\boxed{P(A) = 64.3\%}$$

c-

Let event B: P(red or yellow)

$$P(B) = \frac{n(B)}{n(S)} = \frac{\binom{3}{0}\binom{6}{1}\binom{5}{0} + \binom{3}{0}\binom{6}{0}\binom{5}{1}}{\binom{14}{1}}$$

$$P(B) = \frac{(1)(6)(1) + (1)(1)(5)}{14}$$

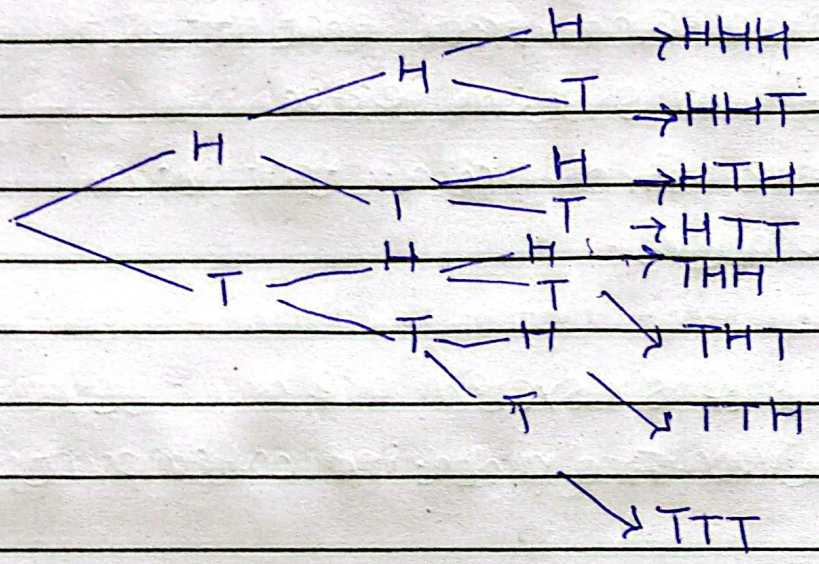
Date: _____ Day: _____

$P(B) = 11/14 = 0.786$

$\boxed{P(B) = 78.6\%}$.

**Example:**

A Fair coin is tossed three times. Find Prob of sequence Head, Tail, Heads




Let event A: P(H, T, H)

$P(A) = \frac{n(A)}{n(S)} = \frac{1}{8} = 0.125$

$\boxed{P(A) = 12.5\%}$

Date: _____ Day: _____

## Example:

Box car contain six complex electronic system. Two of six are to be randomly selected through testing and then classified as defective or non defective.

a) Two of six system are actually defective. Find proba that atleast one of two system tested will be defective. Find proba: that both are defective.

b) Four of six system are actually defective. Find proba: indicated part (a)

a:

Given Data:

Total System = 6

Defective = 2

Non-Defective = 4

Select = 2

(1) Let event A: P(atleast one is Defective)

$$P(A) = \frac{n(A)}{n(S)} = \frac{\binom{2}{1}\binom{4}{1} + \binom{2}{2}\binom{4}{0}}{\binom{6}{2}}$$

$$P(A) = \frac{(2)(4) + (1)(1)}{15} = \frac{9}{15}$$

Date: Day:

$P(A) = 0.6$

$\boxed{P(A) = 60\%}$

(2) Let event B: P(two\Both are defective)

$$P(B) = \frac{n(B)}{n(S)} = \frac{\binom{2}{2}\binom{4}{0}}{\binom{6}{2}} = \frac{(1)(1)}{15}$$

$P(B) = 1/15 = 0.067$

$\boxed{P(B) = 6.7\%}$

b-

Given Data:

Total System = 6

Defective = 4 Select = 2.

Non-Defec = 2

(1) Let event A: P(atleast one is Defec)

$$P(A) = \frac{n(B)}{n(S)} = \frac{\binom{4}{1}\binom{2}{1} + \binom{4}{2}\binom{2}{0}}{\binom{6}{2}}$$

$$P(A) = \frac{8+6}{15} = \frac{14}{15} = 0.933$$

$\boxed{P(A) = 93.3\%}$

Date: Day:

(2) Let event B : P(Both defective)

$$P(B) = \frac{n(B)}{n(S)} = \frac{\binom{4}{2}\binom{2}{0}}{15} = \frac{(6)(1)}{15} = \frac{6}{15} = 0.4$$

$$\boxed{P(B) = 40\%}$$

**Example:**

In a tank containing 10 Fishes, there are three Yellow and seven black fishes. We select three fishes at random.

a) Prob that exactly one Yellow Fish selected
b) Prob that at most one Yellow Fish
c) Prob that atleast one Yellow Fish

Given Data:

Total Fish = 10 , select Fish = 3

Yellow = 3 , seven = 7

a-

Let event A : P(one Yellow Fish)

$$P(A) = \frac{n(A)}{n(S)} = \frac{\binom{3}{1}\binom{7}{2}}{\binom{10}{3}} = \frac{(3)(21)}{120}$$

$$P(A) = \frac{63}{120} = 0.525$$

Date: _____ Day: _____

$$\boxed{P(A) = 52.5\%}$$

b-

Let event B: P(atmost one Yellow Fish)

$$P(B) = \frac{n(B)}{n(S)} = \frac{\binom{3}{1}\binom{7}{2} + \binom{3}{0}\binom{7}{3}}{\binom{10}{3}}$$

$$P(B) = \frac{(3)(21) + (1)(35)}{120} = \frac{98}{120}$$

$$P(B) = 0.817$$

$$\boxed{P(B) = 18.7\%}$$

c-

Let event C: P(atLeast one Yellow)

$$P(C) = \frac{n(C)}{n(S)} = \frac{\binom{3}{1}\binom{7}{2} + \binom{3}{2}\binom{7}{1} + \binom{3}{3}\binom{7}{0}}{\binom{10}{3}}$$

$$P(C) = \frac{(3)(21) + (3)(7) + (1)(1)}{120} = \frac{63 + 21 + 1}{120}$$

$$P(C) = \frac{85}{120} = 0.708$$

$$\boxed{P(C) = 70.8\%}$$

Date: Day:

## Example:

From containing 15 white, 7 black, and 8 yellow balls a sample of 3 ball is Drawn at random. Find the prob that

a) ALL three are yellow
b) ALL three are of same color.
c) ALL three are of diff. color.

Given Data:

Total Balls = 30 black = 7
yellow = 8 white = 15
Select = 3 $n(S) = \binom{30}{3}$

### a:

Let event A: P(ALL are yellow)

$$P(A) = \frac{n(A)}{n(S)} = \frac{\binom{15}{0}\binom{7}{0}\binom{8}{3}}{\binom{30}{3}}$$

$$P(A) = \frac{(1)(1)(56)}{4060} = 0.014$$

$$\boxed{P(A) = 1.4\%}$$

Date: _____ Day: _____

b-

Let event B: P(three are of ~~the~~ same color)

$$P(B) = \frac{n(B)}{n(S)} =$$

$$P(B) = \frac{\binom{15}{3}\binom{7}{0}\binom{8}{0} + \binom{15}{0}\binom{7}{3}\binom{8}{0} + \binom{15}{0}\binom{7}{0}\binom{8}{3}}{\binom{30}{3}}$$

$$P(B) = \frac{(455)(1)(1) + (1)(35)(1) + (1)(1)(56)}{4060}$$

$$P(B) = \frac{546}{4060} = 0.134$$

$$\boxed{P(B) = 13.4\%}$$

c-

Let event, C: P(three are of Diff colors)

$$P(C) = \frac{n(C)}{n(S)} = \frac{\binom{15}{1}\binom{7}{1}\binom{8}{1}}{\binom{30}{3}}$$

$$P(C) = \frac{(15)(7)(8)}{4060} = \frac{840}{4060}$$

$$P(C) = 0.207$$

$$\boxed{P(C) = 20.7\%}$$